

TAMPINES JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION



CANDIDATE
NAME

--

CIVICS GROUP

1	2			
---	---	--	--	--

TUTOR
NAME

--

Biology

9648/03

Applications Paper and Planning Question

Friday, 20 September 2013

Paper 3

2 hours

Candidates answer on the Question Paper

Additional Materials are required: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination:

1. Fasten all answer scripts for Question 5 securely;
2. Hand in Paper 3 and the answer scripts for Question 5 **separately**.

For Examiner's Use		
1	15	
2	15	
3	10	
4	12	
5	20	
Total	72	

The number of marks is given in brackets [] at the end of each question or part question.

STRUCTURED QUESTIONS (40 MARKS)

Answer **all** the questions in this section in the space provided.

QUESTION ONE (15 MARKS)

One of the applications of RFLP analysis is for restriction mapping. *Bam*H1, *Eco*RI and *Hind*III restriction enzymes are used to map a 15kb bacterial DNA. Three reactions were set up to cut the 15kb DNA sample, each with one of the three enzymes. After digestion with restriction enzymes, the DNA fragments were separated into bands. Results of the reactions are shown in **Fig 1.1**.

Describe briefly how, in such DNA analysis,

- (i) DNA is separated into bands; [4]

- (ii) the bands are made visible. [1]

- (iii) Based on the information given and the results in **Fig 1.1**, which of the following map (W, X, Y or Z) in **Fig 1.2** is not a possible map? [1]

Map _____

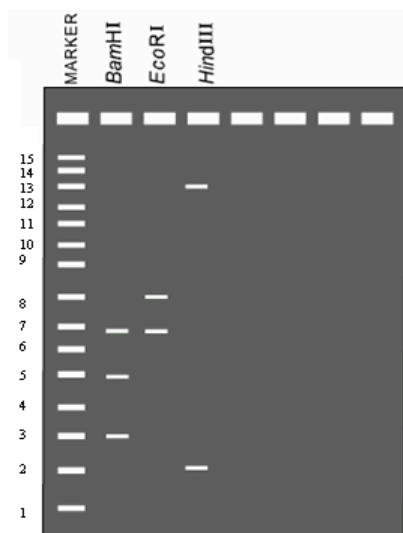


Fig 1.1

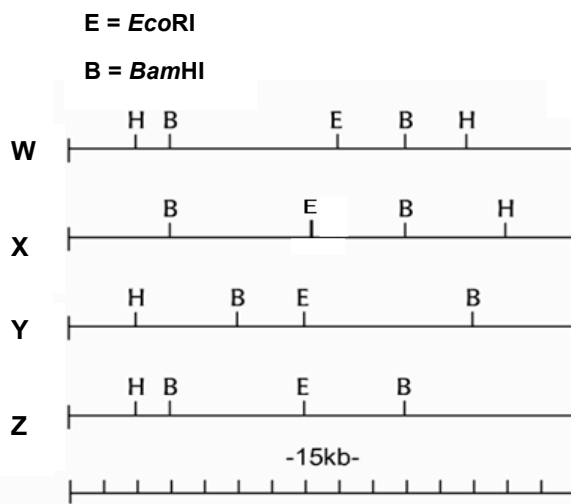


Fig 2.2

Scale

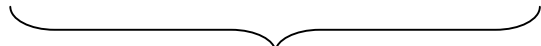


Fig 1.2

15kb

[Turn Over]

Further procedures were performed by setting up additional digestions using pairs of restriction enzymes: *Bam*H1/*Eco*R1, *Bam*H1/*Hind*III, and *Eco*R1/*Hind*III in three separate reactions.

- (iv) With reference to the results shown in **Fig 1.3**, explain which of the following W, X Y or Z is the correct map. [3]

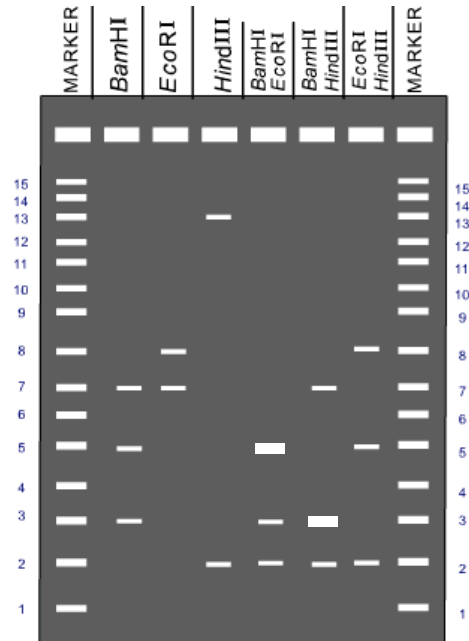
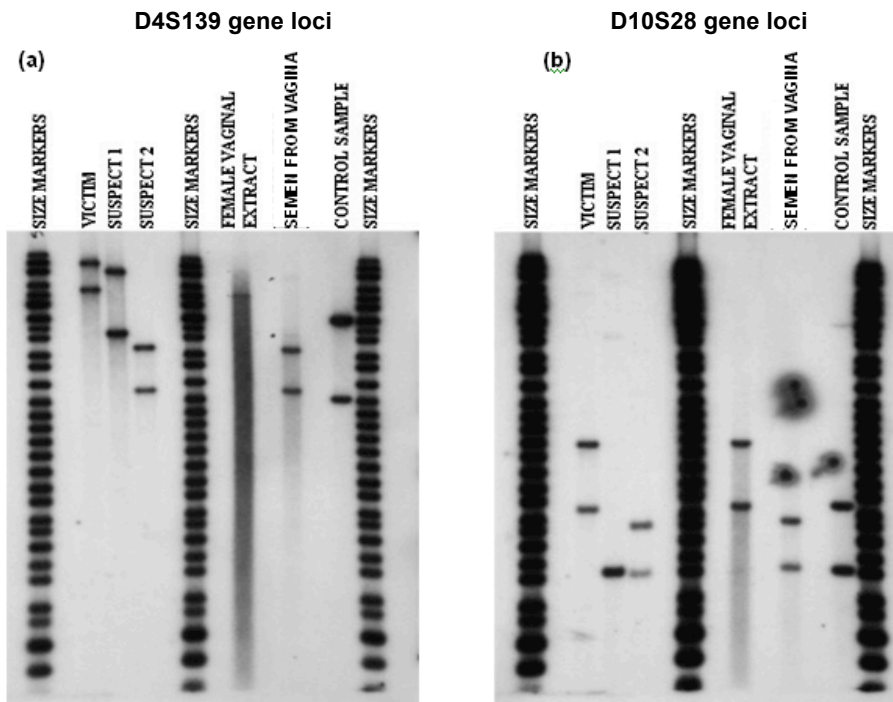


Fig 1.3

Another application of RFLP analysis is for forensic scientist to identify criminals. Two men (Suspect 1 and 2) were accused of kidnapping and raping a woman. RFLP analysis was used to examine the genetic characteristics of the two suspects, the victim, and semen from the vagina of victim. Two genetic loci, D4S139 (a) and D10S28 (b), were examined. These two loci are variable number of tandem repeats (VNTR).

Fig 1.4 shows two autoradiograms of the DNA banding patterns. Three lanes of the autoradiograms (on the far left, far right, and middle) display multiple bands called size markers.



(i) State the function of the size markers. [2]

(ii) Explain the term VNTR. [1]

(iii) Identify and explain the criminal with evidences shown in **Fig 1.4**. [3]

[Total: /15 marks]

QUESTION TWO (15 MARKS)

Children with severe combined immunodeficiency disorder (SCID) cannot produce the many types of white blood cells that fight infections. This is because they do not have the functional gene to make the enzyme, adenosine deaminase (ADA).

Researchers are currently exploring the possibility of using stem cells in therapy for SCID. Some children with SCID have been treated with stem cells. Stem cells can divide and develop into any type of blood cell, including white blood cells.

The treatment used with the children is described in **Fig. 2.1**.

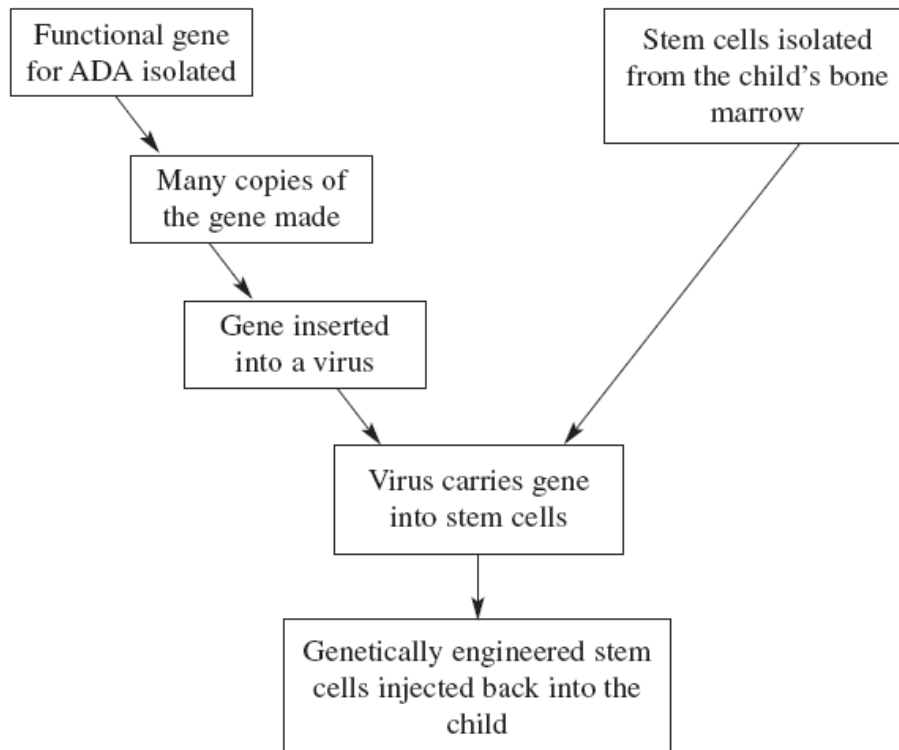


Fig. 2.1

- (a) Using the information given, state **two** reasons why stem cells were used in this treatment. [2]

- (b) Suggest the type and the potency of stem cells that were used in this treatment. [2]

Type: _____

Potency: _____

A child was treated with genetically engineered stem cells. **Fig. 2.2** shows the number of functioning white blood cells in the child during the year following treatment. Children who do not suffer from SCID have between 5000 and 8000 white blood cells per mm^3 of blood.

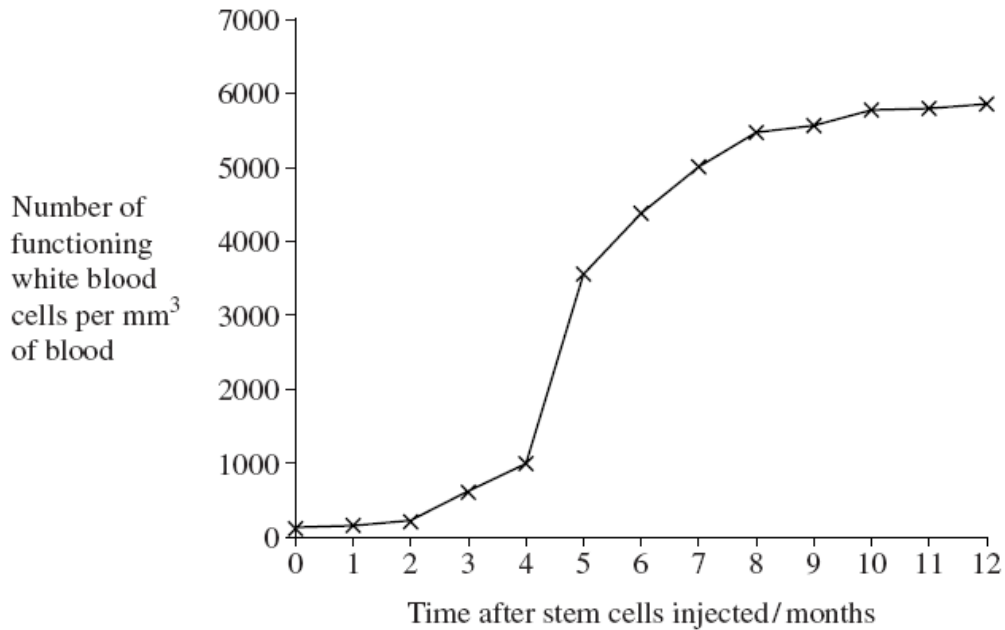


Fig. 2.2

(c) Describe the results shown in **Fig. 2.2**. [3]

(d) Explain why a person treated by gene therapy may still have children who suffer from SCID. [2]

SCID is an inherited autosomal recessive disorder. Therefore it is easier to treat SCID by gene therapy.

- (e) Suggest why it is easier to perform gene therapy when a mutant allele is recessive instead of dominant. [1]

Although the use of gene therapy for treatment of genetic disorders in humans seems promising, there are still many concerns to be addressed.

- (f) Explain the factors that may keep gene therapy from becoming an effective treatment for genetic disorders. [3]

- (g) Discuss **one** ethical and **one** social concern regarding the use of gene therapy. [2]

[Total: /15 marks]

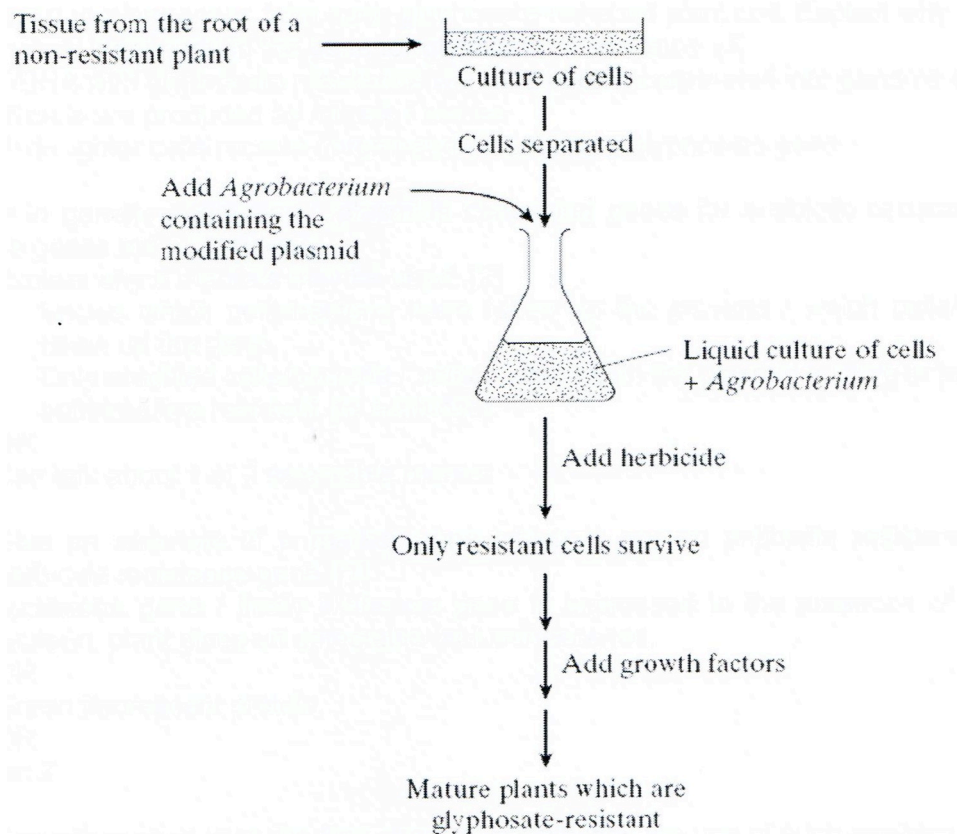
QUESTION THREE (10 MARKS)

Read the following passage.

Genetic engineering is a technique for improving crop plants. It is a process by which a single new gene can be added to the genes already present in the plants. One way of doing this is to use a natural genetic engineer, a soil microorganism called *Agrobacterium*. This bacterium possesses a plasmid which can be modified in the laboratory so that it becomes the carrier of new genetic information.

For example, a gene coding for the ability to break down the herbicide glyphosate can be introduced with the assistance of the enzymes restriction endonuclease and ligase. Once inside the crop plant, the gene for glyphosate breakdown makes the plant resistant to the effects of this herbicide. All surrounding weeds are destroyed when sprayed with glyphosate.

The flow-chart shows how the plasmid carrying this gene can be used to produce glyphosate-resistant plants.



Use the information from the passage and your own knowledge to answer the following questions.

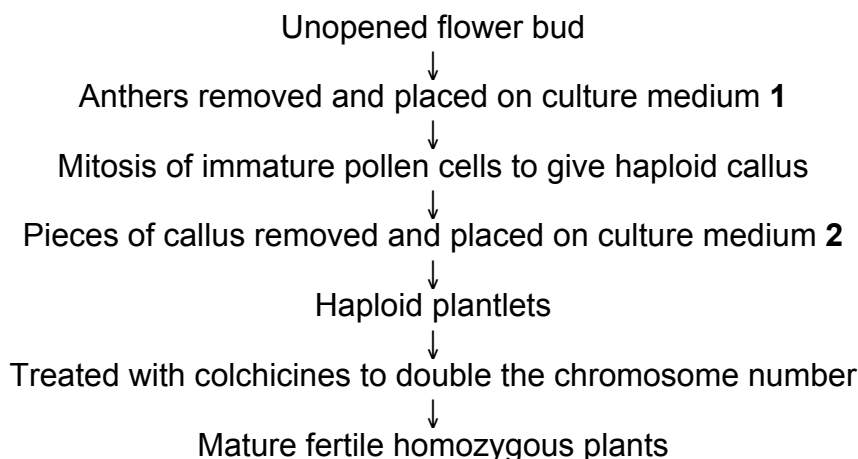
- (a) One mature plant grows from each glyphosate-resistant plant cell. Explain why all the cells of the mature plant contain the gene for glyphosate-resistance. [1]

- (b) Outline the technique of tissue culture needed to obtain matured plants from the transformed cells which are resistant to the herbicide. [3]

- (c) Suggest and explain **one** possible problem associated with the use of herbicides together with genetically modified, herbicide-resistant crop plants. [1]

Another technique required in plant biotechnology is plant tissue culture. Various organs or tissues of a plant may be used in tissue culture to produce a large number of genetically identical plants.

In the process of anther culture, haploid plantlets are produced from immature anthers, from which mature fertile homozygous plants are obtained. The process is shown diagrammatically in the figure below.



- (d) With reference to the figure,
 (i) Explain what is meant by haploid callus. [2]

- (ii) State one precaution that would be taken when transferring material onto culture medium 1 or 2. [1]

- (iii) Suggest why variation may be generated in plantlets derived from the same callus. [1]

- (iv) Suggest how colchicine could bring about the doubling of chromosome number in haploid cells. [1]

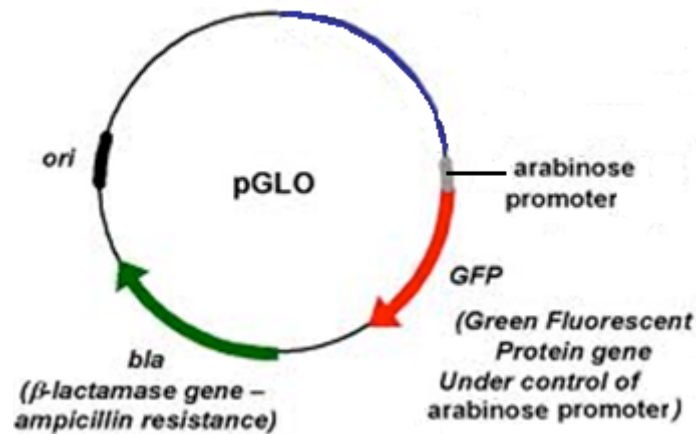
[Total: /10 marks]

PLANNING QUESTION

QUESTION FOUR (12 MARKS)

You are required to plan, but not carry out, an investigation into the transformation of *Escherichia coli* bacteria using the pGLO plasmid as shown below.

The GFP gene is regulated by a promoter of the arabinose operon. The operon can be induced by the presence of arabinose. The successful uptake of the pGLO can be detected by bacteria that fluorescence green under UV light.



(Modified from <http://www.mikeblaber.org/oldwine/BCH4053I/Lecture07/Lecture07.htm>)

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- microfuge tubes
- different types of agar plates
- foam microfuge tube holder
- parafilm sealing tapes
- stopwatch
- thermometer
- micropipettes and tips
- incubator oven
- transformation solution
- plasmid DNA solution (pGLO)
- competent *Escherichia coli* cells in microfuge tube
- sterile water
- tripod stand and bunsen burner
- hot water and crushed ice
- UV visualiser

Your plan should have a clear and helpful structure to include

- an explanation of theory to support your practical procedure
- a description of the method used including the scientific reasoning behind the method
- how you will record your results and ensure that they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- the correct use of technical and scientific terms.

FREE-RESPONSE QUESTION (20 MARKS)

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections (a), (b) etc., as indicated in the question.

QUESTION FIVE

- (a) Describe PCR and compare and contrast with replication process. [8]
- (b) Outline the process of gel electrophoresis and explain how it is able to separate fragments of DNA. [9]
- (c) State the uses of RFLP analysis. [3]

End Of Paper